

Learning Dynamics for High-Speed Tractor-Trailer Control

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Abstract—Model Predictive Control (MPC) is commonly employed for effective control of autonomous vehicles, but challenges arise when the dynamics model fails to capture uncertainty. Learning-based MPC has been explored to improve control in uncertain environments, which we study and apply to tractor-trailer (TT) systems. TT system dynamics are especially unstable in reverse driving and depend on difficult-to-measure quantities, such as road friction and suspension dynamics. We present a learned dynamics model applied to TT control, and introduce a reduced-order dynamics model with Fiala tire forces as an analytic baseline. We compare our fully learned dynamics model within an MPPI controller against an analytic prior and several baseline controllers. We find that our fully learned dynamics model offers superior control in simulated environments with different levels of uncertainty over the analytic prior and baseline controllers, allowing for more reliable target velocity tracking and better jackknife avoidance. [\[Project Page\]](#) [\[Code\]](#)

I. INTRODUCTION

Control of autonomous vehicles has seen significant progress in recent years [1], [2], but the problem of effective control in uncertain environments continues to remain an active research question [3]. Model Predictive Control (MPC) is a commonly used framework that employs a dynamics model to optimize a finite-horizon control sequence while accounting for input and state constraints. Consequently, inaccurate or incomplete information about the system dynamics and operating environments can substantially degrade closed-loop MPC performance [4].

These challenges are particularly pronounced in tractor-trailer systems, which are underactuated and nonlinear [5]. The mechanical coupling between the actuated tractor and the passive trailer makes the vehicle combination especially unstable, particularly during reverse motion [6]. Tractor-trailer dynamics are also sensitive to variables that are difficult to measure accurately, including load transfer, road friction, road slope, and suspension dynamics. These properties make the tractor-trailer system a challenging testbed for control under uncertainty.

A. Related Work

Recent advances in the direction of handling dynamical and environmental uncertainty have focused on integrating learned dynamics into MPC controllers, reducing the reliance on a fixed analytic dynamic prior [3], [7]–[11].

One alternative to explicit dynamics learning is policy adaptation: Rapid Motor Adaptation (RMA) adapts a base policy online through a module that infers a latent representation of environmental parameters [12], [13]. Safe DPA [14] augments an adaptive policy with a learned dynamics model and a CBF safety filter. We focus on a different paradigm: we learn the dynamics of the system directly rather than adapting an RL policy online.

Direct learned dynamics are generally implemented as a fully learned predictor [9], [10] or as a residual correction to an analytic prior [15]. Gaussian processes (GP) are commonly used for the learning framework, as they provide native uncertainty quantification output and have been shown to be effective estimators in low-data regimes [11]. More recently, however, neural dynamics have arisen as an alternative [3], [7], [16] because they scale to large offline datasets and can use finite histories of state–action pairs to infer latent operating conditions and mitigate partial observability [9], [10].

Existing methods for tractor-trailer control [17]–[19] focus on parking, navigation, and collision avoidance in constrained environments. When high speeds have been explored, control has instead relied on fixed analytic models and has not addressed reverse driving at handling limits [20], [21]. Learned vehicle dynamics have also been used to improve robustness to state-estimation error and environmental variation, but prior work has focused on single-body vehicles rather than tractor-trailer systems [22]. As such, there is a gap in the literature surrounding the application of learning-based MPC models for tractor-trailer control in uncertain environments at high speeds and in reverse.

B. Contributions

We provide a case study of learned dynamics applied to the high-speed control of tractor-trailer systems under both system and environmental uncertainty. Our main contributions are:

- We find that a learning-based MPPI controller demonstrates superior performance over various baseline controllers for high-speed forward and reverse tractor-trailer control under uncertain vehicle and environmental conditions.
- We provide a controlled comparison between learned and analytic prior dynamics predictors with and without privileged environmental information, characterizing their multi-step prediction divergence over a look-ahead time horizon.
- We derive a reduced-order tractor-trailer dynamics model with Fiala tire forces, providing an analytic

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control baseline.

II. PRELIMINARIES

A. Problem Formulation

Consider a discrete-time nonlinear system

$$\mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{u}_t) \quad (1)$$

where $\mathbf{x}_t \in \mathcal{X} \subseteq \mathbb{R}^n$ is the state at time $t \in \mathbb{N}$, $\mathbf{u}_t \in \mathcal{U} \subseteq \mathbb{R}^m$ is the control input at time t , and $f : \mathcal{X} \times \mathcal{U} \rightarrow \mathcal{X}$ is the discrete-time dynamics.

We assume the full system state $\mathbf{x}$ is only partially observable, so we partition $\mathbf{x}_t = (\mathbf{s}_t, \mathbf{z}_t)$, where $\mathbf{s}_t \in \mathcal{S} \subseteq \mathbb{R}^s$ is observable and $\mathbf{z}_t \in \mathcal{Z} \subseteq \mathbb{R}^{n-s}$ is unobservable. We also assume that the system is controlled through a closed-loop controller, which at every step solves an optimal control problem (OCP).

Precisely, our trajectory optimization OCP at each timestep t can be stated as follows: for a plant state $\mathbf{x}$, fix $\mathbf{x}_0$ as the initial state, $T \in \mathbb{N}$ as a fixed lookahead time, and $C : \mathcal{S} \times \mathcal{U} \rightarrow \mathbb{R}$ as some cost function. We solve the trajectory optimization problem

$$\begin{aligned} \arg \min_{\mathbf{u}_{0:T}} \sum_{t=0}^T C(\mathbf{s}_t, \mathbf{u}_t) \\ \text{s.t. } \mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{u}_t), \quad \mathbf{u}_t \in \mathcal{U}, \quad t = 0, \dots, T-1. \end{aligned} \quad (2)$$

Because f and $\mathbf{z}_t$ are unavailable to the controller, this ideal problem is implemented using an internal model of the observable dynamics.

B. Model Predictive Path Integral Control

Eq. (2) is commonly solved with methods like nonlinear programming; however, Model Predictive Path Integral (MPPI) control is a recent method that proves promising for this objective [7], [23]. MPPI reframes the trajectory optimization problem as importance sampling from a distribution $\mathcal{N}(\hat{\mathbf{u}}_{0:T}, \Sigma)$ around some prior control sequence $\hat{\mathbf{u}}_{0:T}$.

At each iteration, MPPI performs K forward simulation sequences or “rollouts” for sampled control sequences $\mathbf{v}_{0:T} \sim \mathcal{N}(\hat{\mathbf{u}}_{0:T}, \Sigma)$ under an internal dynamics function $\tilde{f} : \mathcal{S} \times \mathcal{U} \rightarrow \mathcal{S}$. For each rollout, the horizon index t is reset to $t = 0$ at the current controller time, with $\tilde{\mathbf{s}}_0^{(k)} = \mathbf{s}_t$. Denoting the predicted state-action pair for the k -th rollout index at time t as $(\tilde{\mathbf{s}}_t^{(k)}, \mathbf{v}_t^{(k)})$, the k -th rollout’s state prediction for time t is calculated as

$$\tilde{\mathbf{s}}_t^{(k)} = \tilde{f}(\tilde{\mathbf{s}}_{t-1}^{(k)}, \mathbf{v}_{t-1}^{(k)}). \quad (3)$$

Then, the rollout’s cost is recovered as

$$S^{(k)} = \sum_{t=1}^T \left[c(\tilde{\mathbf{s}}_t^{(k)}, \mathbf{v}_t^{(k)}) + \gamma \hat{\mathbf{u}}_t^\top \Sigma^{-1} (\mathbf{v}_t^{(k)} - \hat{\mathbf{u}}_t) \right], \quad (4)$$

where the second term is a control deviation cost from the reference with γ as a tunable parameter. Then, the sampling

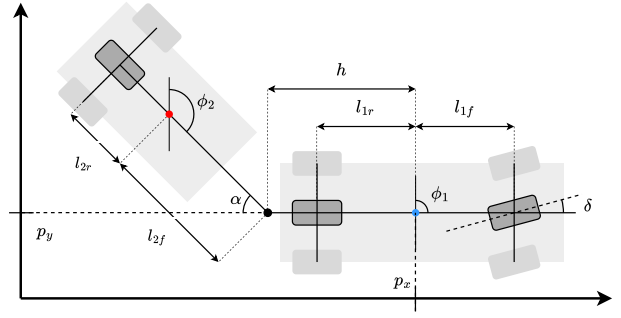


Fig. 1: Illustration of our tractor-trailer model. Employs bicycle model geometry with an additional wheel for the trailer, which is rigidly connected to the tractor at the hitch point.

weight is calculated as

$$\eta = \sum_{k=1}^K \exp \left(-\frac{1}{\lambda} \left(S^{(k)} - \min_j S^{(j)} \right) \right) \quad (5)$$

$$w^{(k)} = \frac{1}{\eta} \exp \left(-\frac{1}{\lambda} \left(S^{(k)} - \min_j S^{(j)} \right) \right) \quad (6)$$

where λ is a tunable parameter representing the inverse temperature.

C. Neural dynamics

As the MPPI controller relies on the internal dynamics representation $\tilde{f} : \mathcal{S} \times \mathcal{U} \rightarrow \mathcal{S}$ for rollout and importance weighting, it is imperative for $\tilde{f}$ to accurately represent the plant dynamics f . In particular, a single observation at time t may not adequately represent complete environment and system conditions. Therefore, as in [9], instead of interpreting the system as a Markovian $\mathbf{s}_{t+1} = \tilde{f}(\mathbf{s}_t, \mathbf{u}_t)$, we allow a fixed horizon of length H of past observed states and controls as input, leading to a function $\mathbf{s}_{t+1} = \tilde{f}(\mathbf{s}_{t-H+1}, \mathbf{u}_{t-H+1}, \dots, \mathbf{s}_t, \mathbf{u}_t)$. We implement $\tilde{f}$ as a multilayer perceptron (MLP).

III. METHODOLOGY

In this section, we present our analytic tractor-trailer dynamics model with Fiala tire forces, describe our learned dynamics model (full and residual dynamics prediction) and how it integrates with MPPI, and detail the simulation environment used to evaluate the controller under various uncertainty sources.

A. Tractor-trailer Dynamics

We derive an analytic tractor-trailer dynamics model in two dimensions with a configuration similar to that of [24], though we do not assume the longitudinal accelerations are constant. We adopt the Fiala brush tire model in a similar manner to [25]. This model serves as our baseline dynamics.

We assume a bicycle geometry configuration of the trailer, as shown in Fig. 1, with the trailer modeled as a single rear wheel attached to the hitch. Any quantity subscripted with 1 refers to the tractor, while 2 refers to the trailer. Therefore, ϕ_1, ϕ_2 are the yaws of the tractor and trailer, respectively; l_{1r}, l_{2r} are the distances from the tractor and the trailer’s

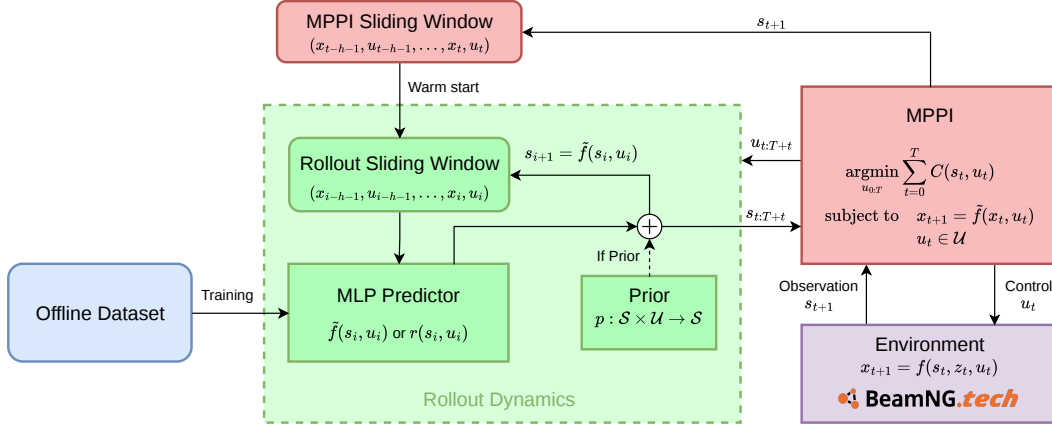


Fig. 2: Overview of our controller architecture. The MPC controller uses either the fully learned or residual dynamics model to compute trajectory rollouts. The learned dynamics MLP predicts dynamics from a sliding window history of state-action pairs and is trained from an offline dataset of collected trajectories.

center of mass to the modeled rear wheel; and l_{1f}, l_{2f} refer to the distance from the center of mass of the tractor and trailer to the modeled front wheel and the hitch point, respectively. Moreover, h is the distance between the tractor's center of mass and the hitch point, $\alpha := \phi_1 - \phi_2$ is the hitch angle, and δ is the steering angle.

We define $\mathbf{q} := (p_{1x}, p_{1y}, \phi_1, \phi_2, v_{1x}, v_{1y}, \dot{\phi}_1, \dot{\phi}_2)$, where the angles are defined as before and $\mathbf{p}_1 := (p_{1x}, p_{1y})$ are the x, y coordinates of the tractor, and v_{1x}, v_{1y} are the longitudinal and lateral tractor velocities. In the same convention, we define $\mathbf{f}_{1f} := (f_{1xf}, f_{1yf})$ as the tractor's front tire forces in the body frame, $\mathbf{f}_{1r} := (f_{1xr}, f_{1yr})$ as the tractor's rear tire forces in the body frame, and $\mathbf{f}_{2r} := (f_{2xr}, f_{2yr})$ as the trailer's rear tire forces.

Let $c := \cos \alpha$, $s := \sin \alpha$, $\bar{m} := m_1 + m_2$, $\rho := m_2 l_{2f}$, and let R_θ denote rotation by θ , so that R_α maps tractor-frame components to trailer-frame components. Body-frame velocities carry the subscript of their body, and $\mathbf{v} := (v_{1x}, v_{1y}, \dot{\phi}_1, \dot{\phi}_2)$.

Let $\mathbf{f}_h$ be the force the trailer exerts on the tractor at the hitch, expressed in the tractor frame. Newton-Euler in each body frame gives

$$\begin{aligned} m_1 (\dot{\mathbf{v}}_1 + \dot{\phi}_1 J \mathbf{v}_1) &= \mathbf{f}_{1r} + R_\delta \mathbf{f}_{1f} + \mathbf{f}_h \\ I_{z1} \ddot{\phi}_1 &= -l_{1r} f_{1yr} + l_{1f} (R_\delta \mathbf{f}_{1f})_y - h f_{hy} \quad (7) \\ m_2 (\dot{\mathbf{v}}_2 + \dot{\phi}_2 J \mathbf{v}_2) &= \mathbf{f}_{2r} - R_\alpha \mathbf{f}_h \\ I_{z2} \ddot{\phi}_2 &= -l_{2r} f_{2yr} - l_{2f} (R_\alpha \mathbf{f}_h)_y, \quad (8) \end{aligned}$$

with $J := \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. The hitch is rigid, so the trailer velocity is determined by the tractor state and $\dot{\phi}_2$,

$$\mathbf{v}_2 = R_\alpha \begin{bmatrix} v_{1x} \\ v_{1y} - h \dot{\phi}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ -l_{2f} \dot{\phi}_2 \end{bmatrix}. \quad (9)$$

Differentiating (9) yields

$$\dot{\mathbf{v}}_2 = R_\alpha \begin{bmatrix} \dot{v}_{1x} \\ \dot{v}_{1y} - h \ddot{\phi}_1 \end{bmatrix} + \dot{\alpha} J R_\alpha \begin{bmatrix} v_{1x} \\ v_{1y} - h \dot{\phi}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ -l_{2f} \ddot{\phi}_2 \end{bmatrix}. \quad (10)$$

Note that Eq. (7) and Eq. (8) cannot be solved sequentially: $\dot{\mathbf{v}}_1$ depends on $\mathbf{f}_h$, which by (8) depends on $\dot{\mathbf{v}}_2$, which by (10) depends on $\dot{\mathbf{v}}_1$.

Then, we substitute (9), (10) into (8) to eliminate $\mathbf{f}_h$, and collecting the four unknowns yields a system of the form

$$M(\alpha) \dot{\mathbf{v}} = \mathbf{b}_{\text{tire}} + \mathbf{b}_{\text{cor}}, \quad (11)$$

where

$$M(\alpha) = \begin{bmatrix} \bar{m} & 0 & 0 & -\rho s \\ 0 & \bar{m} & -m_2 h & -\rho c \\ 0 & -m_2 h & I_{z1} + m_2 h^2 & \rho h c \\ -\rho s & -\rho c & \rho h c & I_{z2} + m_2 l_{2f}^2 \end{bmatrix}. \quad (12)$$

Writing $\mathbf{f}_1 := \mathbf{f}_{1r} + R_\delta \mathbf{f}_{1f}$ for the total tractor tire force, $\hat{\mathbf{f}}_2 := R_\alpha^\top \mathbf{f}_{2r}$ and $\tilde{\mathbf{v}}_2 := R_\alpha^\top \mathbf{v}_2$ for the trailer force and velocity expressed in the tractor frame, and $g := \rho \dot{\alpha} \dot{\phi}_2$, we have

$$\mathbf{b}_{\text{tire}} = \begin{bmatrix} f_{1x} + \hat{f}_{2x} \\ f_{1y} + \hat{f}_{2y} \\ -l_{1r} f_{1yr} + l_{1f} (R_\delta \mathbf{f}_{1f})_y - h \hat{f}_{2y} \\ -(l_{2f} + l_{2r}) f_{2yr} \end{bmatrix}, \quad (13)$$

$$\mathbf{b}_{\text{cor}} = \begin{bmatrix} \dot{\phi}_1 (m_1 v_{1y} + m_2 \tilde{v}_{2y}) + g c \\ -\dot{\phi}_1 (m_1 v_{1x} + m_2 \tilde{v}_{2x}) - g s \\ m_2 h \dot{\phi}_1 \tilde{v}_{2x} + h g s \\ \rho v_{2x} \dot{\phi}_1 \end{bmatrix}. \quad (14)$$

Lateral tire forces are modeled using the Fiala brush model,

$$f_y = \begin{cases} -C_c \tan \beta + \frac{C_c^2 |\tan \beta| \tan \beta}{3\mu f_z} - \frac{C_c^3 \tan^3 \beta}{27\mu^2 f_z^2}, & |\beta| < \beta_{sl}, \\ -\mu f_z \operatorname{sgn} \beta, & \text{otherwise,} \end{cases} \quad (15)$$

where C_c denotes the cornering stiffness, $\beta_{sl} = \arctan(3\mu f_z / C_c)$ denotes the slip angle at full saturation and $\beta = -\operatorname{atan2}(v_\perp, |v_\parallel|)$ denotes the slip angle of each wheel. The use of $|v_\parallel|$ ensures that f_y opposes the lateral slip velocity $v_\perp$ regardless of the direction of travel.

Then, the dynamics are obtained by inverting (11) directly to $\dot{\mathbf{v}} = M^{-1}(\mathbf{b}_{\text{tire}} + \mathbf{b}_{\text{cor}})$.

B. Controller Architecture

We give an overview of the controller pipeline, which is illustrated in Fig. 2.

1) *Dataset Collection*: For our experiment, an offline dataset of trajectories is first collected through MPPI with the analytic model to control the plant. Then, with the dataset, we train an MLP to predict the dynamics. Trajectories are collected across different friction levels and target velocities. Noise is injected into the control inputs. Ideally, the offline dataset would contain a large spread of possible driving behaviors to best represent the full dynamics of the vehicle. We augment our dataset by collecting additional forward driving trajectories that were driven in a sinusoidal weaving pattern generated using an augmented cost function designed to better excite the tire forces. We employ some iterative augmentation of the dataset, where the MLP then collects more data at handling limits to add to the dataset.

2) *Model Training*: We adopt a dynamics prediction format where the model predicts the full dynamics of the system $\tilde{f}$ given a fixed-size state history. The dynamics MLP consist of four hidden layers with 128 units each and take in a state-action pair history of $H = 4$ time steps. The model takes input

$$(\sin \alpha, \cos \alpha, v_x, v_y, \dot{\phi}_1, \dot{\phi}_2, \delta, a, \hat{\delta}, \hat{a})$$

and outputs

$$(\dot{v}_x, \dot{v}_y, \ddot{\phi}_1, \ddot{\phi}_2, \hat{\delta}, \hat{a})$$

for the next time step, where $\hat{\delta}, \hat{a}$ are the plant steering and acceleration states respectively, and δ, a are the commanded steering and acceleration from the controller. In the case of residual dynamics, the MLP predicts the residual for the first four components from the analytic prior, and the realized control inputs are still fully learned. Dynamic quantities are then integrated with standard explicit Euler. In training, all data were subject to z -score standardization.

3) *Controller Overview*: As depicted in Fig. 2, during MPPI rollouts, predicted state-action pairs from the learned dynamics MLP are fed through the sliding window. Then, given the K predicted rollouts, the MPPI controller calculates the optimal control input, which is then given to the plant. At this point, a new state observation is obtained, which is concatenated with its corresponding control input to form the next state-action pair. This is then fed back into the sliding window, completing the cycle and warm-starting the MLP predictor for the next iteration. In implementation, the sliding window is initially populated using H iterations of the analytic MPPI controller. Otherwise, undefined behavior could result, as artificial warm starts could be out of distribution.

4) *Cost Function*: We define a cost function for the controller that promotes constant-velocity, safe driving. To keep the tractor-trailer on the track and to prevent jackknifing, we define constraint violation c_{viol} as

$$c_{\text{viol}} = c_{\text{track}} + c_{\text{jackknife}} \quad (16)$$

$$c_{\text{track}} = \max(0, |e_{\text{lat}}| - 0.45W_{\text{track}} + 0.1)$$

$$c_{\text{jackknife}} = \max(0, |\alpha| - \alpha_{\text{max}})$$

where e_{lat} is the signed deviation of the tractor's center of mass from the centerline of the track, α_{max} is the maximum hitch angle, and W_{track} is the width of the track. When $|\alpha| > \alpha_{\text{max}}$, we consider the tractor-trailer to have jackknifed. Additionally, we define

$$c_{\text{vel}} = |v_{\text{target}} - v_{\text{track}}| \quad (17)$$

$$c_{\text{center}} = e_{\text{lat}}^2 \quad (18)$$

$$c_{\text{hitch}} = |\alpha| \quad (19)$$

where v_{target} is the desired velocity, and v_{track} is the current velocity of the vehicle projected onto the centerline. Then, combining (16)–(19), our cost function C for predicted step t is given by

$$C = 0.99^t \cdot 10^7 c_{\text{viol}} + k_v c_{\text{vel}} + k_c c_{\text{center}} + k_a c_{\text{hitch}}, \quad (20)$$

where k_v , k_c , and k_a are penalty weights.

C. Simulation Environments

We evaluate the controller using the BeamNG.tech [26] plant, whose internal dynamics are unavailable to the controller. For our plant, we have control inputs (δ, a) , and observation outputs $(p_x, p_y, \phi_1, \phi_2, v_{1x}, v_{1y}, \phi_1, \phi_2, \hat{\delta}, \hat{a})$. We simulate a racing scenario, where the tractor-trailer system has to drive at high velocities both forward and in reverse along a predefined racetrack. This demands that the controller operate near the vehicle's handling limit and at high velocities, requiring precise knowledge of the vehicle's dynamics. Specifically, we test with the general track layout of the Circuit de Barcelona-Catalunya. Therefore, the plant dynamics are of the form

$$\mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{u}_t) = f((\mathbf{s}_t, \mathbf{z}_t), \mathbf{u}_t), \quad (21)$$

where $\mathbf{z}_t$ contains unobserved plant states and parameters¹, including vehicle pitch and roll, suspension states, local road friction and roughness, and powertrain behavior.

D. Baseline Controllers

We evaluate our learned-dynamics and analytic-prior controllers against three baselines:

- i) *IPOPT*: This baseline uses IPOPT [27] to solve a nonlinear model predictive control (MPC) problem in a receding-horizon fashion, using direct shooting with the analytic dynamics model. The objective penalizes target-speed error, lateral path error, hitch angle, constraint violations, and control effort over a 60-step horizon.
- ii) *LQR-PID*: This baseline uses LQR [28] for steering control and PID [29] for acceleration control. The LQR controller is based on a linearization of the analytic dynamics model, with gains scheduled according to the longitudinal velocity of the tractor-trailer.
- iii) *MPCC*: This baseline uses the contouring and lag error formulations of [30], replacing the original RC-car dynamics model with our analytic dynamics model.

¹BeamNG includes numerous latent soft-body, suspension, tire, and powertrain states beyond those listed here.

IV. RESULTS

We evaluate the following questions in our experiment:

1. **Q1:** Does the fully learned rollout model demonstrate significant improvement in closed-loop performance relative to the analytic controllers with prior dynamics in a uniform, flat asphalt environment?
2. **Q2:** Does this performance advantage persist under unobserved friction and terrain variations, including slope and banking absent from the training data?
3. **Q1:** How much of the performance advantages of learned dynamics arises from inferring latent environmental parameters versus system parameters?

A. Environment Results

We evaluate forward and reverse tractor-trailer racing behavior in four different scenarios, as shown in Fig. 3. Representative reverse-driving trajectories are shown in Fig. 4.

1. *Asphalt*: Uniform high-friction, flat racing track. The ground is uniformly asphalt. There is no variety in terrain.
2. *Snow*: Varied friction, flat racing track. The ground is split noisily between small snow patches and asphalt. There is no variety in terrain.
3. *Slope*: Uniform high-friction, but the track exhibits elevation changes and banking.
4. *Snow + Slope*: The same asphalt-snow pattern as in the Snow scenario, with the same terrain pattern as the Slope scenario.

We use the controller architecture specified in Fig. 2, with hyperparameters shown in Table I.

TABLE I: MPPI configuration for forward and reverse driving.

Parameter	Forward	Reverse
k_v	1×10^2	5×10^1
k_c	1×10^0	5×10^1
k_a	7×10^2	1×10^2
λ (inv. temp.)	150	100
Σ (steer, accel)	$(7 \times 10^{-2}, 0.2)$	$(2 \times 10^{-2}, 0.2)$
K (samples)	500	500
T (horizon steps)	80	65
γ	0.0	0.0
Δt (step)	0.05	0.05

We now elaborate on the results for the four scenarios, of which a summary can be found in Table II.

1) *Asphalt Scenario*: We observe that the fully learned controller is the only controller that successfully completes all 500 steps without terminating prematurely across velocity setpoints. The fully learned controller is the only controller to complete all nine runs at every velocity setpoint. It achieves the lowest RMS hitch angle at both reverse setpoints. In forward motion, LQR-PID achieves a slightly lower RMS hitch angle, but completes fewer runs. The MPPI utilizing our analytic dynamics is less successful and fails to achieve the given v^* , especially compared to the fully learned model. The performance of the baseline controllers is highly variable across velocities. Each baseline suffers from a high $\bar{\alpha}_{\text{rms}}$ on at least one velocity, even compared to the analytic MPPI, and some also fail to achieve the desired v^* , such as MPCC.

2) *Snow Scenario*: The same trends from the asphalt environment are also observed in the snowy environment. As before, the fully learned controller succeeds in all trials and maintains a $\bar{v}$ near $|v^*|$. Despite traveling at a much higher average speed in reverse than analytic-prior MPPI, the fully learned controller does not jackknife or stall in any trial, whereas analytic-prior MPPI stalls in four of nine runs at each reverse setpoint. The baseline controllers generally complete fewer runs. LQR-PID exhibits higher $\bar{\alpha}_{\text{rms}}$ than the fully learned controller in reverse motion and ends some trials with jackknifing or going off-track, although it achieves a slightly lower $\bar{\alpha}_{\text{rms}}$ in forward motion. The IPOPT-based controller completes fewer simulation steps than the fully learned controller and exhibits substantially larger $\bar{\alpha}_{\text{rms}}$ values at every setpoint. Finally, the MPCC controller completes the fewest simulation steps on average across all controllers tested at each velocity, with the fewest completed runs overall.

3) *Slope Scenario*: We observe similar trends to the asphalt and snow scenarios. While the fully learned controller completes slightly fewer runs on the out-of-distribution slope terrain, it still exhibits stronger velocity tracking and lower $\bar{\alpha}_{\text{rms}}$ than the analytic-prior controller at every setpoint. This suggests that the learned dynamics model transfers effectively to this unseen environment. The fully learned controller also completes more reverse runs and achieves lower reverse $\bar{\alpha}_{\text{rms}}$ than all baseline controllers. In forward motion, however, LQR-PID matches its completion rate while achieving a slightly higher mean speed and lower $\bar{\alpha}_{\text{rms}}$.

4) *Snow + Slope Scenario*: Combining snow and uneven terrain, we observe that the fully learned controller maintains an overall advantage over the other controllers tested, although stalling limits its reverse-driving performance. Notably, it is the only controller that does not fail due to jackknifing in any trial and exhibits lower $\bar{\alpha}_{\text{rms}}$ in nearly all comparisons. These results suggest stronger hitch-angle regulation, particularly during reverse motion. While LQR-PID achieves a higher $\bar{v}$ and slightly lower $\bar{\alpha}_{\text{rms}}$ in the forward-driving experiment, its reverse-driving outcomes are worse, with fewer successful completions and more jackknifing or off-track incidents.

5) *Scenario Test Overview*: In terms of overall performance, the learned model broadly demonstrates an advantage over the analytic prior in all three tested scenarios, along with a superior ability to avoid jackknives, and answers **Q1** and **Q2**. The third and fourth scenarios, “Slope” and “Snow + Slope,” are particularly notable, as slope and banking variation was never present within the learned model’s training data. Unlike [16], the learned model is not provided information about the environment’s terrain, which injects additional uncertainty, adding to the difficulty of this task.

We compare how the error of the learned model and the prior changes from the second to the third and fourth scenario. In Table II, both N and $\bar{v}$ exhibit a significant decline for the Fiala prior, whereas the learned model decays less noticeably. From this trend, we see that the model is able to predict dynamics more accurately even when exposed to

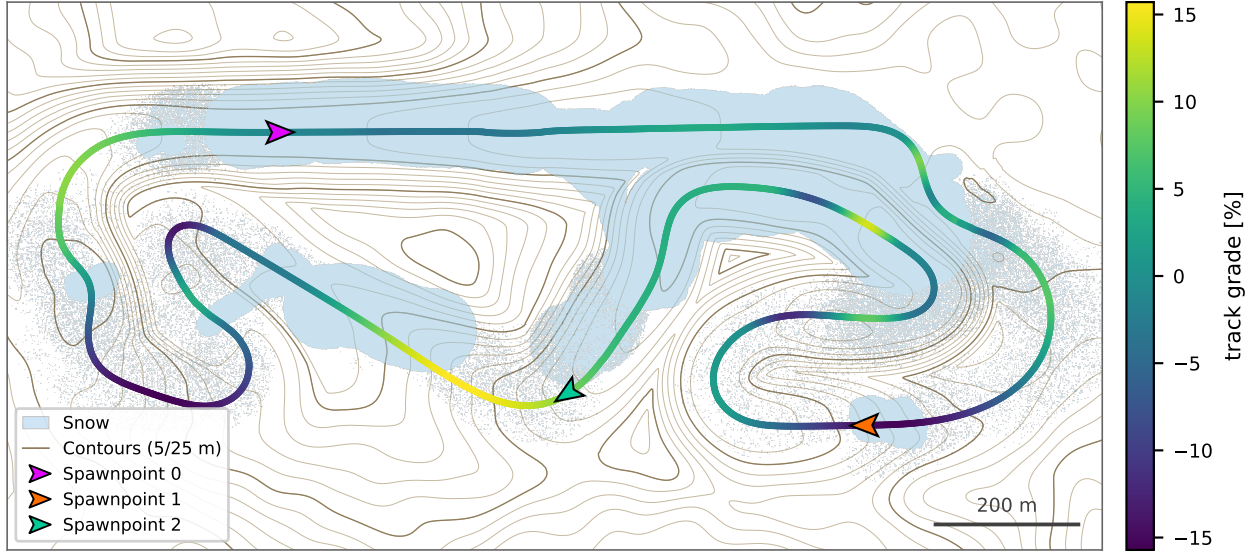


Fig. 3: Top-down view of the “Snow + Slope” testing environment, which follows the layout of the Barcelona-Catalunya racetrack with terrain and ground material modifications. Snow and terrain modifications are additive on the base asphalt environment. To ensure testing consistency, the same snow pattern is used in the “Snow” case and the “Snow + Slope” case. Similarly, the same terrain is used in the “Slope” case and the “Snow + Slope” case.

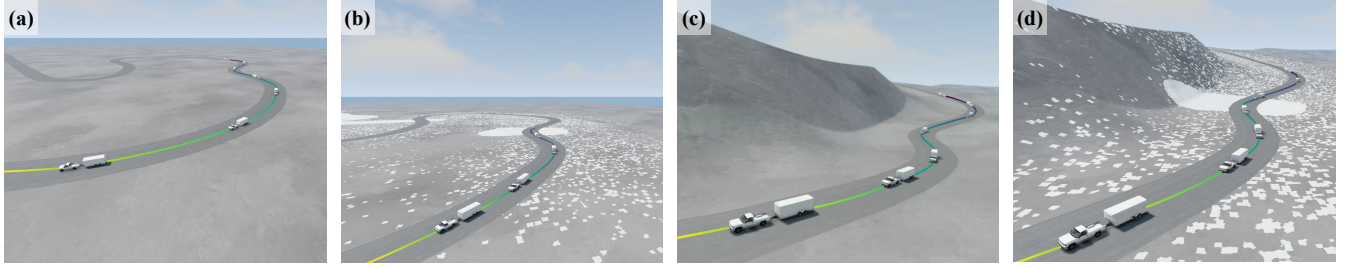


Fig. 4: Simulated reverse trailer driving path for our 3 scenarios [(a): Asphalt, (b): Snow, (c): Slope, (d): Snow + slope] under fully learned dynamics control. White spots denote snowy, low-friction regions. The gradient line denotes the vehicle’s trajectory over time.

TABLE II: Closed-loop tracking across four environments, with nine runs per configuration.

Controller / rollout model	v^* (km/h)	Asphalt				Snow				Slope				Snow + slope			
		$\bar{v}$ (km/h)	$\bar{\alpha}_{rms}$ (deg)	$\bar{N}$ (steps)	C/J/S/O/F (out of 9)	$\bar{v}$ (km/h)	$\bar{\alpha}_{rms}$ (deg)	$\bar{N}$ (steps)	C/J/S/O/F (out of 9)	$\bar{v}$ (km/h)	$\bar{\alpha}_{rms}$ (deg)	$\bar{N}$ (steps)	C/J/S/O/F (out of 9)	$\bar{v}$ (km/h)	$\bar{\alpha}_{rms}$ (deg)	$\bar{N}$ (steps)	C/J/S/O/F (out of 9)
Fully learned	−30	29.1	1.14	500	9/0/0/0/0	29.1	1.18	500	9/0/0/0/0	25.4	1.03	455	8/0/1/0/0	20.4	10.82	386	6/0/3/0/0
	−50	47.2	1.75	500	9/0/0/0/0	46.0	1.82	500	9/0/0/0/0	42.1	3.06	498	8/1/0/0/0	30.4	15.41	413	4/0/5/0/0
	80	75.4	2.57	500	9/0/0/0/0	72.6	2.72	500	9/0/0/0/0	70.7	2.67	500	9/0/0/0/0	69.1	2.83	500	9/0/0/0/0
Fiala prior	−30	19.7	5.25	392	5/0/4/0/0	19.6	7.82	403	5/0/4/0/0	22.3	2.86	498	8/0/1/0/0	16.7	11.47	365	4/0/5/0/0
	−50	22.7	6.70	448	6/0/3/0/0	22.8	8.49	417	5/0/4/0/0	22.5	6.48	457	6/0/3/0/0	18.1	15.86	321	2/1/6/0/0
	80	71.1	3.00	500	9/0/0/0/0	68.7	3.26	500	9/0/0/0/0	67.1	3.19	500	9/0/0/0/0	64.8	3.66	500	9/0/0/0/0
LQR-PID	−30	30.8	4.80	495	8/0/0/1/0	30.6	5.36	439	6/0/0/3/0	28.5	10.02	446	6/0/0/3/0	24.0	16.56	377	3/6/0/0/0
	−50	45.1	14.41	452	6/1/0/2/0	48.8	8.97	407	6/3/0/0/0	46.9	10.58	396	6/3/0/0/0	37.1	16.91	352	3/3/0/3/0
	80	74.8	2.45	436	6/0/0/3/0	72.4	2.36	442	6/0/0/3/0	71.7	2.52	500	9/0/0/0/0	70.7	2.53	500	9/0/0/0/0
IPOPT	−30	28.4	3.11	500	9/0/0/0/0	23.5	20.22	373	5/1/3/0/0	19.9	16.32	478	7/2/0/0/0	16.7	20.79	338	4/0/5/0/0
	−50	32.2	54.93	107	0/9/0/0/0	31.9	39.96	244	0/6/0/3/0	26.3	28.09	293	3/3/3/0/0	23.5	30.59	292	3/3/3/0/0
	80	71.3	12.28	471	5/1/0/3/0	69.1	9.16	468	6/3/0/0/0	69.5	8.13	490	6/0/0/3/0	66.7	11.80	475	6/0/0/3/0
MPCC	−30	22.0	26.57	301	0/3/2/4/0	21.9	39.16	261	0/3/3/3/0	18.1	30.46	374	1/0/7/0/1	14.1	31.79	275	0/1/6/2/0
	−50	39.3	14.74	261.3	0/3/1/4/1	30.2	24.95	203	0/2/3/4/0	37.3	17.98	352	1/3/1/3/1	25.5	25.63	202	0/1/3/5/0
	80	68.0	6.06	400	4/0/0/5/0	66.1	4.82	407	4/0/0/4/1	64.5	6.73	379	3/0/0/5/1	64.4	5.74	426	4/0/0/5/0

Each configuration comprises three seeds at each of three locations ($n = 9$). Our notation is as follows: v^* is the signed velocity setpoint; negative values denote reverse motion. $\bar{v}$ and $\bar{\alpha}_{rms}$ are the arithmetic means of the per-run mean speed magnitudes and RMS hitch angles, respectively, computed after the first 100 control steps. $\bar{N}$ is the mean number of completed control steps out of 500, with $\Delta t = 0.05$ s. All means include unsuccessful runs and weight each run equally. **Bold** entries indicate that all nine runs completed 500 steps. C/J/S/O/F gives the numbers of completed, jackknifed, stalled, off-track, and solver-failed runs, respectively; dividing each count by nine gives its empirical rate. Completion denotes reaching the step limit. A stall is defined by $|v_x| < 0.5$ m/s with no track progress for 2 s.

a hidden variable not found in its training data, pointing to a broader general understanding of dynamics. In practical

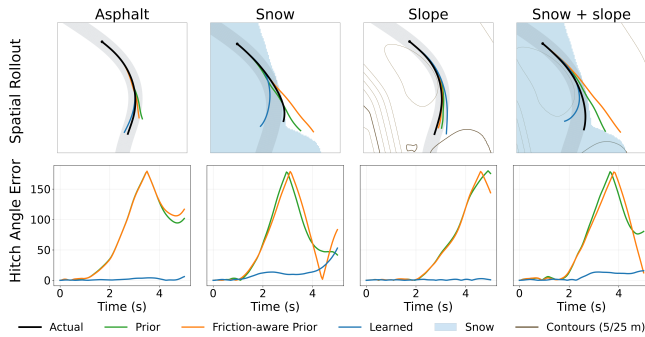


Fig. 5: Trajectory rollout errors by environment and dynamics model in forwards driving.

terms, it is potentially unnecessary to gather a dataset that covers all possible vehicle behaviors in all environmental contexts to achieve a generalized understanding of the dynamics.

B. Dynamics Prediction Error

We compare the multi-step dynamics prediction accuracy of the following models:

1. The analytical prior with no environmental friction information; we fix $\mu = 1$.
2. The analytical prior with environmental friction lookup.
3. The fully-learned model with $H = 4$.

From an actual state-action history of forward and reverse driving simulated in BeamNG with the learned dynamics MPPI controller, we compare the open-loop 5 second rollout prediction divergence of our candidate dynamics models, shown in Fig. 5, 6, Table III.

For backwards driving, we observe that the learned model generally holds an advantage to both the prior and friction-aware prior dynamics. Visually, from Fig. 6, we can see that the hitch angle error of the learned model increases more gradually through time compared to the others across all four environments. As a result, we conclude that the learned model provides more accurate hitch predictions, critical for maintaining control and avoiding jackknifing.

For forwards driving, the learned model is also superior to the other models in rollout accuracy. In Table III, the learned model achieves the lowest hitch angle error of all the models, and Fig. 5 shows that it maintains a much more stable error. We note that the spikes in the hitch angle error in the other models are consistent with the loss of control of the trailer with it spinning around, resulting in the apparent dip in error later in the rollout.

V. CONCLUSION

In this work, we evaluated learned dynamics for MPPI-based forward and reverse tractor-trailer control. We compared a fully learned model with a residual model constructed from a Fiala-based analytic prior. In open-loop evaluation, the learned model achieved superior performance both in position and hitch angle tracking, as well as better scaling through time relative to the prior and μ -aware prior models. The learned model controller achieved the target velocities

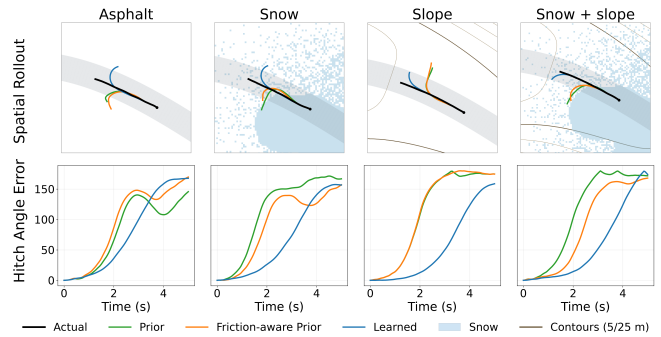


Fig. 6: Trajectory rollout errors by environment and dynamics model in reverse driving.

TABLE III: Position (p_{rms}) and hitch-angle (h_{rms}) RMSE across environments. Bold values indicate the lowest RMSE for each environment, direction, and metric.

Direction	Environment	Prior		μ -aware Prior		Learned	
		p_{rms}	h_{rms}	p_{rms}	h_{rms}	p_{rms}	h_{rms}
Backward	Asphalt	5.8	96.5	7.4	113.0	4.2	101.1
	Snow	7.0	130.1	4.9	107.9	4.0	88.2
	Slope	7.7	132.6	8.2	134.2	2.2	78.7
	Snow + Slope	8.1	131.5	7.3	111.6	2.1	83.4
Forward	Asphalt	7.3	93.4	6.6	95.3	1.4	2.1
	Snow	7.3	82.3	11.9	83.8	10.8	16.0
	Slope	3.5	80.4	2.7	79.2	5.7	1.1
	Snow + Slope	8.4	85.7	14.7	84.0	7.5	8.7

more reliably than the residual model controller, especially in reverse driving experiments, and exhibited generally lower predicted hitch angle error, indicating better jackknife avoidance and stability. Notably, this remained true even when terrain with sloped roads was introduced, which was not included in its training data, suggesting that the learned model possesses a more generalizable representation of the system dynamics than the residual model. Future work should explore other sources of uncertainty, such as trailer payload mass, to further improve robustness.

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